

Custom Stack Vs Off-the-shelf Counterparts

Off-the-shelf	Custom
Large code size/memory footprint (limits memory available for application layer)	Small code size of around 16KB for the MAC and Stack combo (the MAC is typically around 2KB and the Stack is around 14KB)
Slow performance on 8-bit microprocessors	It performs well on inexpensive 8-bit microprocessors
Large RAM requirements for repeaters/routers	Less RAM requirement
Requires a lot of expertise to create the application layer	A custom MAC on its own is an incomplete solution. As discussed, the MAC is the portion of the system that interfaces the PHY to the Stack, so it is also necessary to develop a Custom Stack and an Application Layer. It also requires a massive amount of IEEE 802.15.4 PHY knowledge
Requires a lot of expertise to configure and manage the network	It requires a massive amount of specialised networking software knowledge and expertise
Throughput is limited	It runs fast with low latency and high throughput
Power consumption is relatively high	Its power consumption is reasonable
It includes support for Star, Tree, and Mesh topologies	Proprietary point-point links. Proprietary Star
It includes support for encryption—AES-128 and CCM	No standard security service is provided
It supports high-level protocols that facilitate interoperability.	It is not interoperable
It is an open standard. Product require certification to ensure compatibility	It is free. No certification is required. It is affordable even if it isn't free
It can take nine months (or more) to develop and test a new solution	It can take one-to-two years (or more) to develop and test a new solution

RF signals. Stack refers to the software that defines the communication protocols and the data/control packets that are to be passed around the network. MAC layer is the software that interfaces the stack to the PHY. Finally, the application layer will monitor signals from any sensor and control the actuators.

To reduce time during the development and deployment of the application layer, it is always advisable to use off-the-shelf solutions (Fig. 7), e.g., OpenMAC (802.15.4 MAC) from Meshnetics, SimpliciTI (802.15.4) and Z-Stack from TI, SNAP from Synapse or 802.15.4 MAC, and ZigBee Network layer from Silicon Laboratories. But a custom network can also be created from ground up as shown in Fig 7. This usually involves licensing and modifying an existing, simple MAC to create a custom MAC. A custom stack and the application layer are created later.

The following features therefore need to be considered while choosing an off-the-shelf communication software stack or creating custom MACs and stacks for a wireless sensor product:

1. The level of standard service provided
2. Primary types of applications

3. Memory size (RAM/Flash)
4. Cost
5. Efforts required to develop application software
6. Security
7. Channel power scan (CCA, ED, and LQI)
8. Effective data throughput
9. Whether it supports external RF active components (PA and LNA)
10. Whether it supports RF testing
11. Support for low power modes
12. Whether available as ANSI C source code
13. Ease of porting to other MCUs

Case studies

Over the last few years, the IEEE 802.15.4 standard and ZigBee have created a stir in the electronics world due to the benefits they offer compared to other wireless standards available today. They are cheap to implement, optimised for very low power operation, and outstanding in terms of reliability and portability.

IEEE 802.15.4/ZigBee-based WSN.

The key design requirements in any WSN depend on long battery life, low cost, a small footprint, and mesh networking between a large number of devices in an interoperable and multi-application environment. These have driven the specifications for 802.15.4

and ZigBee to provide the network infrastructure required for WSN applications. While IEEE 802.15.4 defines the physical and MAC layers, ZigBee defines the network and application layers stacked on top of it, working in the licence-free 868/915/2.4 GHz frequency bands.

ZigBee network topologies. IEEE 802.15.4 and ZigBee support three different network topologies—star, mesh and cluster tree. Of these, mesh networks have good self-configuration, self-healing, self-routing and self-management capabilities.

ZigBee devices. A ZigBee network can include devices like ZigBee coordinator (ZC), ZigBee router (ZR) and ZigBee end device (ZED).

ZigBee coordinator. It is a device that sets up the network, acts as a portal to manage the network, monitors its performance and configures parameters. Network address allocation, routing table maintenance and management in a ZigBee network need large amounts of computation. As networks expand and applications become more complex, the burden on the network coordinator becomes heavier. Therefore a high-performance controller with more powerful computing and processing capabilities must be employed. In large systems,